

Play the Uno from an external MIDI controller

For: musicians and testers using QuarkWave's Uno R4 sound module without the Pico, or alongside it. **Status:** source-reviewed, gateway smoke-tested, physical DIN hardware untested (firmware snapshot: 2026-09-23). The physical DIN connector is unbuilt; its [proposed isolated input](#) is not an as-built connection.

Choose an input

Input	What this build supports	What to check on the assembled instrument
Separate DIN MIDI	The Uno listens for MIDI through a software serial receiver on pin 2. It accepts all MIDI channels.	Confirm the installed DIN connector and interface circuit before connecting a controller. Source pin assignments alone are not a wiring guide.
RTP-MIDI	Enabled when the optional Pico is attached. The Pico hosts the <code>QuarkWave</code> session on UDP port <code>5004</code> and forwards MIDI to the Uno.	Connect to the Pico's network address or advertised session. The Uno alone has no wireless MIDI in this build.
BLE-MIDI	Disabled in this build.	Do not expect a Bluetooth MIDI device to discover QuarkWave.

[Uno MIDI instances and build flags](#), [Pico network gateway](#), [physical connection notes](#).

Play without the Pico

1. Power the assembled Uno and connect its tested audio output to your listening setup at a low level. Connect your controller through the build's verified DIN input.
2. Send Note On and Note Off messages. **Expected from source:** the Uno plays its built-in startup sound and remains playable without a Pico or browser. This startup sound is the Uno's firmware default, **not** the Pico's Warm Pad factory preset. [Uno defaults](#), [external input handlers](#).
3. Try pitch bend (fixed ± 2 semitones), sustain pedal CC64, or a sound CC from the [technical CC table](#). These are live changes in the Uno; the Uno does not save patch files. [Uno bend and CC handlers](#).
4. Release notes when finished. If a note remains on, send All Notes Off CC123 or All Sound Off CC120 from a controller that supports it, then stop and inspect the MIDI stream. These messages are described by the Uno's CC handler; their behavior needs hardware confirmation. [CC map](#).

Play over Wi-Fi with the Pico attached

1. Power both boards and wait until the browser reports **Uno: Connected to Pico**. Keep the controller and Pico on the same reachable network.
2. In your RTP-MIDI software, connect to the Pico's `QuarkWave` session on UDP port `5004`. If discovery does not show it, use the Pico's network address. For exact Mac and Windows steps, see [Network MIDI setup](#).

3. Connect one wireless controller at a time. If it disconnects while notes or its sustain pedal are held, the Pico releases both on the Uno. The sound follows its release setting; a pedal held from the browser or DIN remains in effect. A silent network loss may take until the RTP session times out to be recognized; use **Panic** if you need an immediate release.
4. Send a note, then release it. **Expected from source:** the Pico forwards both messages over its wired MIDI link. Its first sound message marks external activity so a clean Uno startup does not automatically replace a wireless-controlled sound. The message and disconnect paths have passed a two-board diagnostic; audible output remains untested. [Pico gateway](#), [Uno receiver](#).

Use an external controller while the Pico is present

An external note, CC, bend, or accepted QuarkWave sound command marks standalone activity. If that happens **before** the Pico's boot patch is applied, the Pico keeps the Uno's current sound and the browser shows **Uno: Connected to Pico · standalone sound**. The browser's **Sync Pico patch to Uno** explicitly resets and replaces that sound and held notes. Once a Pico patch has been applied, external MIDI can still play and adjust the Uno, but those changes may not appear in the Pico's patch controls or saved file. [Sync rule](#), [musician takeover steps](#).

Follow MIDI Clock

1. In the Pico's Explore view, enable **Use external MIDI clock**; or send QuarkWave Program Change `100` to the Uno. Clock pulses alone do not select external tempo. Program Change `101` selects internal tempo. Some controller menus display program numbers one higher than the MIDI data byte; verify the transmitted byte with a MIDI monitor. [Tempo commands](#).
2. Set your controller or sequencer to send MIDI Clock and, for transport-controlled arpeggiation, Start, Continue, and Stop. Hold notes with the arpeggiator enabled. **Expected from source:** Start begins at the first step, Continue resumes, and Stop releases the current arp note while retaining held keys. Direct notes remain playable when the arpeggiator is off. [Clock behavior](#).
3. Test at 40–240 BPM. When both DIN and Pico-forwarded RTP send valid recent clock, DIN has priority. After two seconds without pulses from the selected source, the Uno switches to the other recent source or holds the last valid tempo. It starts at 120 BPM before receiving a valid clock. These timing outcomes are hardware untested. [Clock implementation](#).

QuarkWave-specific sound commands

External DIN and RTP devices may send the documented sound Program Changes `100–103`, `110–112`, and SysEx commands `01–03`; see the [exact byte table](#). Configure a controller to transmit those bytes if you need tempo source, delay sync, vibrato waveform, internal BPM, LFO routing, or chorus depth. Factory-preset selection is a Pico patch operation; generic Program Change numbers do not select the Pico's eight factory presets. Visualization, reset, and handshake commands are rejected from RTP peers; the Pico sends them only for its own UI and link management. QuarkWave does not forward its custom messages to other synths. [DIN handlers](#), [Pico RTP gateway](#).